# Diamond chips, from wafer to qubit

ADAMAS: an open, tested framework for diamond electronics and room-temperature quantum processors

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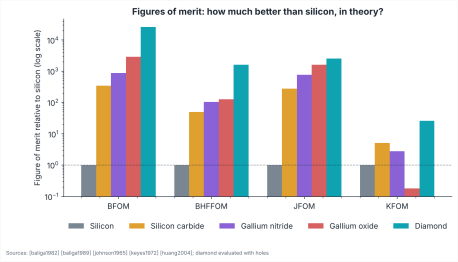
Every number on these slides is generated by the package (`make talk`)

California State University, Dominguez Hills · github.com/Normansrule/adamas-diamond-framework

# Why diamond at all?

- Power-switch figure of merit: 25,777× silicon
- Heat removal: 14.7× silicon
- A defect, the nitrogen-vacancy (NV) center, is a qubit at room temperature

**But:** boron dopants are 0.37 eV deep (under 1% active at 300 K), wafers only reached 3 inches recently, and logic is p-channel only.



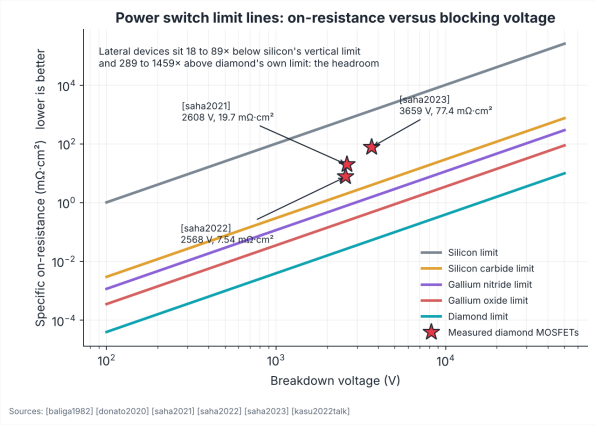
# What ADAMAS is

One connected, executable chain, each link checked against published data or an independent tool:

materials → process → circuits → qubits → error correction → machine size → experiments

- 579 references, 500 verified against Crossref
- 53 figures regenerated from code
- Validation matrix: 18 checks, 17 pass
- Process traveler, PDK, placed 4-bit processor
- Circuit-level error correction (Stim, PyMatching)
- Experiments from US$100, course, 22 web pages

# Power switches: diamond wins when hot



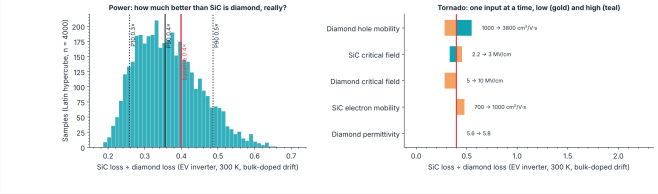

Best measured: 3659 V; 2022 lateral device about 89× below silicon's vertical limit.

Bulk-doped drift: only 0.50% of the boron ionized at 300 K.

| W per switch | 300 K | 523 K |
|---|---|---|
| Silicon | 209.5 | – |
| SiC | 29.7 | 58.0 |
| GaN | 20.6 | – |
| Diamond, bulk | 74.6 | 29.8 |

800 V, 300 A, 20 kHz EV inverter

# But how sure is that? Uncertainty analysis

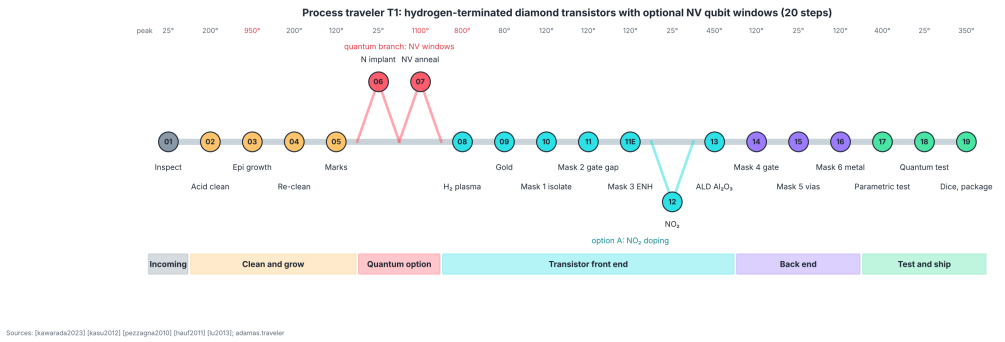



Diamond's critical field (5 to 10 MV/cm) and hole mobility (1,000 to 3,800, baseline 2,000) are not settled.

SiC ÷ diamond loss at 300 K: median 0.36, P10 to P90 0.26 to 0.49; at 523 K: median 1.7×.

Mobility explains 0.64 of the spread, the field 0.17: measure mobility first (a Hall setup, about US$3k).

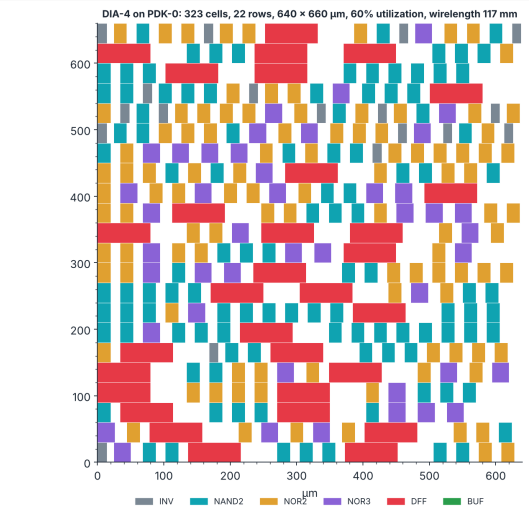
# How you would actually build it




Sources: [kawarada2023] [kasu2012] [pezzagna2010] [hauf2011] [lu2013]; adamas.traveler

The channel is made by surface chemistry (hydrogen termination), gold protects it, one oxygen plasma isolates every device, and all heat is spent before the first metal. Qubit wafers: hydrogen below 500 °C only (hot plasma darkens NV centers).

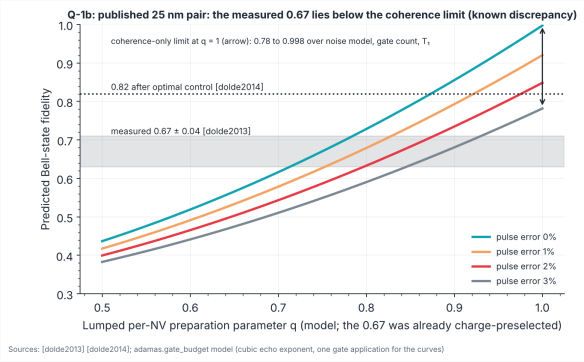
# A processor on diamond




Sources: [weste2011] [meadconway1980] [liu2017] [faggin1996] [ajayi2019]; circuits/digital/place.py

- PDK-0: layers, rules, GDS monitor die, standard cells, LEF, timed Liberty
- DIA-4 (4-bit, CALL/RET) synthesizes to 323 cells, about 1 $\mathrm{mm}^2$ with pads
- Browser emulator matches the Verilog on random programs; ring oscillator within 10% of SPICE

# Two-qubit gates: an honest discrepancy

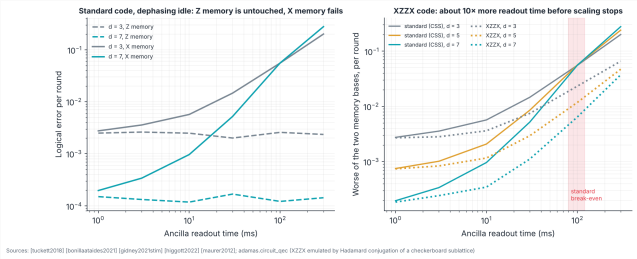



Published room-temperature pair (25 nm, 25 µs gate): coherence alone allows 0.78 to 0.998.

Measured 0.67 (already charge-preselected) lies below the whole range.

The earlier “preparation” explanation is withdrawn; the gap is a known discrepancy. 96% at 9.5 nm has since been shown [joas2025].

# Error correction when readout is slow



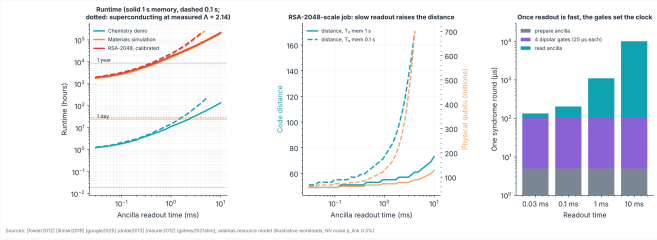

Readout takes about 1 ms while data qubits idle.

$\Lambda = 4.03$ (0.1 ms), 3.56 (1 ms), 2.64 (10 ms).

Scaling stops near 100 ms; the bias-tailored XZZX code pushes that to about 1000 ms.

Native circuits, bias on CZ only (CNOT $\leq 5$): threshold 0.93% $\rightarrow$ 1.81% at $\eta = 100$ (standard code: 0.86%).

# How big, how slow




Sources: [fowler2012] [litinski2019] [google2025] [skoric2023] [maurer2012] [gidney2021stim], adamas resource model (illustrative workloads, NV noise $p_{link}$ 0.3%)

RSA-2048, calibrated to the published 8 h: distance 55, 73 M qubits on a 5.5 mm die.

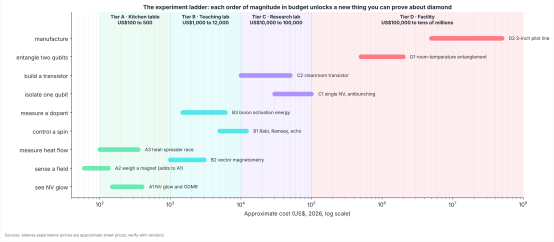
Runtime 2.1 years at 1 ms readout (0.3 at 0.1 ms), P10 to P90 0.4 to 5.3 years; superconducting at today's $\Lambda$: 30 hours.

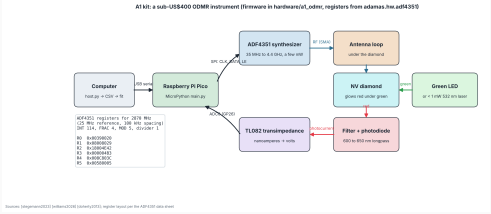
Assumes a 25 µs, 0.3% inter-cell link (not yet demonstrated).

Native circuits: biasing the gates moves the standard code from 135 to 173 M qubits and XZZX from 142 to 58 M.

Area is free; time is not.

# Anyone can start: experiments from US$100




A1 kit: Pico firmware drives an ADF4351 through the 2.87 GHz resonance; data go straight into the fitter.

# Takeaways

1. Diamond power switches win when hot (1.9× less loss than SiC at 523 K) and a bulk-doped switch loses at room temperature: incomplete ionization matters.
2. Room-temperature NV gates measure below what coherence allows, and the gap is unexplained; addressing qubits 10–30 nm apart is the manufacturing problem.
3. A room-temperature quantum computer would be small but slow; readout speed and bias-tailored codes are the levers.
4. Everything is open, tested, and regenerable: `pytest`, `make figures paper talk`.

github.com/Normansrule/adamas-diamond-framework ·
normansrule.github.io/adamas-diamond-framework